

Summary and Recommendations

This analysis evaluates customer churn behavior across **7,043 customers**, examining demographic attributes, service usage patterns, contract types, tenure, and billing characteristics. After data preprocessing (handling missing values, correcting data types, and transforming categorical variables), multiple visualizations were developed to uncover key churn drivers and quantify risk segments.

1. Overall Churn Overview

- The overall churn rate stands at **26.5%**, meaning **1 in every 4 customers (≈1,869 customers)** has discontinued service.
 - Conversely, **73.5% of customers remain retained**, indicating a strong base but a significant improvement opportunity.
 - A reduction of churn by even **5% could improve retention stability and long-term revenue predictability**.
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2. Demographic Insights

Gender

- Churn rates across gender segments are relatively balanced.
- This indicates **gender is not a primary driver of churn**, and retention strategies should focus on behavioral and contract-related factors rather than demographic gender segmentation.

Senior Citizens

- Senior citizens show a **higher churn proportion compared to non-senior customers**.
 - This suggests that elderly customers may experience service complexity, billing confusion, or support dissatisfaction.
 - Targeted communication and simplified service support could potentially reduce churn within this segment.
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3. Tenure-Based Analysis (Strongest Churn Indicator)

Tenure shows the most significant impact on churn:

- Customers with **0–2 months tenure exhibit the highest churn rate**, forming the largest portion of churned customers.
- Customers with **tenure above 24 months show significantly lower churn rates**, indicating increased loyalty over time.
- Churn probability sharply decreases after the first year of service.

Insight:

The **first 3–6 months represent the highest risk period**, making onboarding and early engagement crucial.

4. Contract Type Analysis

- Customers on **month-to-month contracts show significantly higher churn rates** compared to those on longer contracts.
- Customers on **one-year and two-year contracts demonstrate much lower churn percentages.**

Interpretation:

Short-term contracts reduce switching barriers, making customers more likely to leave.
Incentivizing long-term contracts could directly lower churn risk.

5. Payment & Billing Patterns

- Customers using certain electronic payment methods show comparatively higher churn.
- Customers with **higher monthly charges** are slightly more prone to churn, possibly due to price sensitivity.
- However, long-tenure customers often have higher total charges, reflecting customer lifetime value accumulation rather than risk.

Strategic Implication:

Price-value perception plays an important role in churn decisions.

6. Service Adoption & Engagement

- Customers with fewer service subscriptions are more likely to churn.
- Customers engaged with multiple services tend to have higher retention.

This indicates:

Higher product integration increases switching costs and loyalty.

Key Quantitative Highlights

Metric	Value
Total Customers	7,043
Total Churned Customers	~1,869
Overall Churn Rate	26.5%
Retention Rate	73.5%

Metric	Value
Highest Risk Segment	0–2 Month Tenure
High-Risk Contract Type	Month-to-Month

Business Impact & Strategic Recommendations

1 Focus on Early Lifecycle Retention

- Implement structured onboarding programs.
- Provide proactive support within the first 90 days.
- Introduce welcome incentives or loyalty bonuses.

2 Promote Long-Term Contracts

- Offer discounts for 1-year and 2-year commitments.
- Bundle services for improved value perception.

3 Target High-Risk Segments

- Senior citizens
- Month-to-month subscribers
- Low-engagement customers

4 Strengthen Value Communication

- Improve transparency in billing.
 - Emphasize bundled service benefits.
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Strategic Conclusion

The analysis clearly demonstrates that churn is not random — it is concentrated within specific behavioral and contract-driven segments.

With a **26.5% churn rate**, there is substantial opportunity for optimization. By prioritizing:

- Early-stage engagement
- Contract conversion strategies
- Senior customer support enhancement
- Multi-service bundling

The organization can potentially reduce churn by **3–7%**, which would significantly increase long-term customer lifetime value and revenue stability.
